

EvidenceRank — Candidate Proof Engine

- INDIA RUNS Track 1
- CPU-only, offline candidate ranking for approximately 100,000 profiles
- Built for evidence-backed, explainable, and reproducible selection

The problem

- Traditional résumé matching trusts self-reported keywords
- High keyword overlap can hide weak or missing career proof
- Recruiters need reliable top-10 candidates, not merely a large shortlist

Why keyword matching fails

- Keyword stuffing can outrank genuine experience
- Hidden gems may describe strong work in plain language
- Skill claims are not connected to projects, deployment, or evaluation
- No protection exists against contradictory or suspicious profiles

Core idea: prove the match

- Create a compact fingerprint for every candidate
- Separate claimed skills from career evidence
- Rank with proof, production depth, evaluation, risk, and constraints
- Explain every top result using fields already present in the profile

System architecture

- Streaming profiler keeps full-dataset memory use bounded
- JD parsing and proof graphs remain fully local
- Deep risk and calibration work runs only on bounded top pools
- See ``architecture_diagram.mmd`` for the end-to-end flow

Candidate Proof Graph

- Supported: direct title, project, achievement, or career evidence
- Weakly supported: broad profile or education mention
- Unsupported: claim exists without matching evidence
- Short snippets show exactly what the system found

Honeypot Firewall

- Detects keyword stuffing, zero-duration expertise, and contradictions
- Assigns explainable low, medium, high, or severe risk
- Applies bounded penalties instead of aggressive blanket deletion
- Protects the top 10 with stricter deterministic checks

Evidence Calibration

- Rewards retrieval, ranking, evaluation, and production proof
- Checks JD-specific positive and negative constraints
- Uses hireability signals modestly; missing behavior stays neutral
- Calibration cannot replace weak technical evidence

Evaluation and safety

- Offline benchmark: 100% pass rate
- Four-variant ablation compares keyword, baseline, risk, and calibration
- Eight synthetic benchmark cases test intended ranking behavior
- Weight sensitivity checks that results are not tied to one fragile setup
- Final submission safety: PASSED

Submission outputs

- `ranked\_candidates.csv` is the final answer
- `score\_breakdown.csv` exposes component scores and penalties
- Proof, honeypot, calibration, runtime, and reproducibility audits are included
- Submission and demo ZIPs use explicit safe allowlists

Performance and reproducibility

- Measured ranking for 5,000 fingerprints: 9.62 seconds
- Projected ranking for 100,000 fingerprints: 192 seconds
- Observed peak RSS: 57.60 MB
- CPU-only, offline, and independent of external APIs.

Impact

- Recruiters receive evidence-backed rankings instead of keyword counts
- Top candidates come with concise proof and transparent concerns
- Risk-aware reranking improves confidence in the most visible positions
- Human review remains the final hiring decision